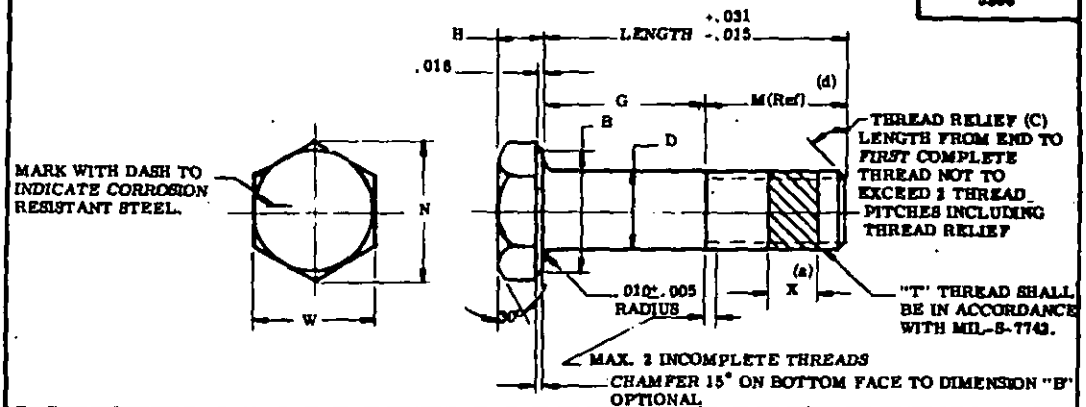


DASH NO.	"T" (b) THREAD UNF-3A	D DIA. MAX.	D DIA. MIN.	B (f) DIA. MIN.	H MAX.	H MIN.	M Ref	N MIN.	W MAX.	W MIN.	DBL. SHEAR STRENGTH POUNDS MIN.	(e) ULT TENSILE STRENGTH POUNDS MIN.
-1	#6-32	.138	.135	.213	.106	.094	.312	.277	.251	.243	1,420	720
-2	#8-32	.164	.161	.275	.118	.106	.437	.348	.313	.305	2,030	1,120
-3	#10-32	.189	.186	.338	.141	.109	.469	.416	.377	.365	2,720	1,600
-4	1/4-28	.249	.246	.398	.172	.140	.506	.488	.440	.428	4,700	2,910
-5	5/16-24	.312	.309	.460	.204	.172	.531	.558	.502	.490	7,370	4,640
-6	3/8-24	.374	.371	.523	.235	.203	.641	.630	.565	.553	10,600	7,020
-7	7/16-20	.437	.433	.585	.266	.234	.656	.701	.627	.615	14,400	9,490
-8	1/2-20	.499	.495	.710	.297	.265	.781	.843	.752	.740	18,800	12,790
-9	9/16-18	.562	.558	.835	.328	.296	.906	.986	.877	.865	23,900	16,250
-10	5/8-18	.624	.620	.898	.360	.328	.953	1.058	.940	.928	29,500	19,200
-12	3/4-16	.749	.744	1.023	.422	.390	.969	1.200	1.066	1.053	42,400	28,000
-14	7/8-14	.874	.869	1.210	.485	.453	1.250	1.413	1.253	1.240	57,700	38,200
-16	1-12	.999	.993	1.398	.547	.515	1.375	1.628	1.441	1.428	75,400	49,700
-28	1-1/8-12	1.124	1.118	1.585	.610	.578	1.500	1.841	1.628	1.615	94,400	64,200
-20	1-1/4-12	1.249	1.243	1.772	.672	.640	1.687	2.054	1.815	1.802	116,500	80,500

(a) LOCATION OF EFFECTIVE LOCKING AREA X SHALL BE IN ACCORDANCE WITH MS15981.
(b) THREADS SHALL BE FORMED BY ANY SINGLE ROLLING PROCESS AND IN ACCORDANCE WITH SPECIFICATION MIL-S-7742.
(c) CHAMFER 45° ± 5°. CHAMFER TO EXTEND TO ROOT OF THREAD.
(d) REFERENCE DIMENSIONS ARE FOR DESIGN PURPOSES ONLY AND ARE NOT AN INSPECTION OR MANUFACTURING REQUIREMENT.
(e) TENSILE VALUES BASED ON 80 KSI ULTIMATE TENSILE STRENGTH UP TO AND INCLUDING 9/16" DIAMETER AND 75 KSI ULTIMATE TENSILE STRENGTH 5/8" DIAMETER AND LARGER. STRESS AREA = 3.1416 (E/2 - 3H/16)². SEE HANDBOOK H-28.
(f) THE DIAMETER OF THE WASHER FACE SHALL NOT EXCEED THE ACTUAL WIDTH ACROSS FLATS.

MATERIAL: CORROSION RESISTING STEEL THAT WILL PROVIDE IN THE FINISHED BOLT THE STRENGTH AND PERMEABILITY SPECIFIED.
TREATMENT: PASSIVATED (SEE PROCUREMENT SPECIFICATION).
SURFACE TEXTURE: THE SURFACE OF THE SHANK, FILLET AND BEARING AREA SHALL NOT EXCEED 125 MICRO-INCHES. SURFACE TEXTURE SHALL BE IN ACCORDANCE WITH ASA B46.1 (1962) ANSI.
LOCKING ELEMENT: IN ACCORDANCE WITH SPECIFICATION MIL-F-18240. LOCKING ELEMENT AREA MUST PASS THRU A RING GAGE EQUAL TO THE MAXIMUM MAJOR THREAD DIAMETER +.003" FREELY OR WITH FINGER PRESSURE.
MAGNETIC PERMEABILITY: THESE BOLTS HAVE A MAGNETIC PERMEABILITY OF 2.0 MAXIMUM (AIR = 1.0) FOR A FIELD OF H = 200 OERSTEDS USING MAGNETIC INDICATOR PER MIL-I-17214.
DESIGN AND USAGE LIMITATIONS: SEE MS15981.
DIMENSIONS IN INCHES.
EXAMPLE OF PART NUMBER: MS21095-4008 = 1/4-28 BOLT, .500" GRIP LENGTH.

FOR DESIGN FEATURE PURPOSES THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

(A) ENTIRE STANDARD REVISED & REDRAWN

APPROVED 2 NOV 65 REVISED (A) 13 APR 71

P.A. NAVY - AS; Other Cost USAF - 11, ARMY - AV	TITLE: BOLT, SELF-LOCKING, 250° F., CRES, 48 KSI Fsu, 80 KSI Ftu, HEX HEAD	MILITARY STANDARD MS21095
PROCUREMENT SPECIFICATION MIL-B-6812 & MIL-F-18240	SUPERSEDES: (A) MS21095(ASG)	SHEET 1 OF 3

DD FORM 672-1 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE. PLATE NO. 15320

REVIEWER SYMBOLS: NAVY - AS, USAF - 11, ARMY - AV
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FED. SUP CLASS
5306

LENGTH DASH NO. (g)	GRIP G ±.010	LENGTH +.031 -.015						
		6-32	8-32	10-32	1/4-28	5/16-24	3/8-24	7/16-20
001	.062	.375	.468	.468				
002	.125	.437	.531	.531	.594			
003	.187	.500	.593	.593	.656			
004	.250	.562	.656	.656	.719			
006	.375	.687	.781	.781	.844	.906	1.016	
008	.500	.812	.906	.906	.969	1.031	1.141	1.156
010	.625	.937	1.031	1.031	1.094	1.156	1.266	1.281
012	.750	1.062	1.156	1.156	1.219	1.281	1.391	1.406
014	.875	1.187	1.281	1.281	1.344	1.406	1.516	1.531
016	1.000	1.312	1.406	1.406	1.469	1.531	1.641	1.656
018	1.125	1.437	1.531	1.531	1.594	1.656	1.766	1.781
020	1.250	1.562	1.656	1.656	1.719	1.781	1.891	1.906
022	1.375	1.687	1.781	1.781	1.844	1.906	2.016	2.031
024	1.500	1.812	1.906	1.906	1.969	2.031	2.141	2.156
026	1.625	1.937	2.031	2.031	2.094	2.156	2.266	2.281
028	1.750	2.062	2.156	2.156	2.219	2.281	2.391	2.406
030	1.875	2.187	2.281	2.281	2.344	2.406	2.516	2.531
032	2.000	2.312	2.406	2.406	2.469	2.531	2.641	2.656
034	2.125	2.437	2.531	2.531	2.594	2.656	2.766	2.781
036	2.250	2.562	2.656	2.656	2.719	2.781	2.891	2.906
038	2.375	2.687	2.781	2.781	2.844	2.906	3.016	3.031
040	2.500	2.812	2.906	2.906	2.969	3.031	3.141	3.156
042	2.625			3.031	3.094	3.156	3.266	3.281
044	2.750			3.156	3.219	3.281	3.391	3.406
046	2.875			3.281	3.344	3.406	3.516	3.531
048	3.000			3.406	3.469	3.531	3.641	3.656
050	3.125			3.531	3.594	3.656	3.766	3.781
052	3.250			3.656	3.719	3.781	3.891	3.906
054	3.375			3.781	3.844	3.906	4.016	4.031
056	3.500			3.906	3.969	4.031	4.141	4.156
058	3.625			4.031	4.094	4.156	4.266	4.281
060	3.750				4.219	4.281	4.391	4.406
062	3.875				4.344	4.406	4.516	4.531
064	4.000				4.469	4.531	4.641	4.656
066	4.125				4.594	4.656	4.766	4.781
068	4.250				4.719	4.781	4.891	4.906
070	4.375				4.844	4.906	5.016	5.031
072	4.500				4.969	5.031	5.141	5.156
074	4.625					5.156	5.266	5.281
076	4.750					5.281	5.391	5.406
078	4.875					5.406	5.516	5.531
080	5.000					5.531	5.641	5.656
082	5.125					5.656	5.766	5.781
084	5.250					5.781	5.891	5.906
086	5.375					5.906	6.016	6.031
088	5.500					6.031	6.141	6.156
090	5.625							6.281
092	5.750							6.406
094	5.875							6.531
096	6.000							6.656
098	6.125							6.781
100	6.250							6.906
102	6.375							7.031

CONT.

REVIEWER SYMBOLS:
NAVY - AS
USAF - 11
ARMY - AV

USER SYMBOLS:

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APPROVED 2 NOV 65 REVISED (A) FOR CHANGES SEE SHEET 1

P.A. NAVY - AS
Other Cust USAF - 11
ARMY - AV

TITLE
BOLT, SELF-LOCKING, 250° F., CRES, 48 KSI Fsu, 80 KSI Ftu, HEX HEAD

MILITARY STANDARD
MS21095

PROCUREMENT SPECIFICATION
MIL-B-6812 & MIL-F-18240

SUPERSEDES (A) MS21095(ASG)

SHEET 2

DD FORM 672-1 (Coordinated)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

PLATE NO. 15520

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LENGTH DASH NO. (g)	GRIP G ±.010	LENGTH +.031 -.015							
		1/2-20	9/16-18	5/8-18	3/4-16	7/8-14	1-12	1-1/8-12	1-1/4-12
008	.500	1.281							
010	.625	1.406	1.531	1.578					
012	.750	1.531	1.656	1.703	1.719				
014	.875	1.656	1.781	1.828	1.844	2.125			
016	1.000	1.781	1.906	1.953	1.969	2.250	2.375		
018	1.125	1.906	2.031	2.078	2.094	2.375	2.500	2.625	
020	1.250	2.031	2.156	2.203	2.219	2.500	2.625	2.750	2.937
022	1.375	2.156	2.281	2.328	2.344	2.625	2.750	2.875	3.062
024	1.500	2.281	2.406	2.453	2.469	2.750	2.875	3.000	3.187
026	1.625	2.406	2.531	2.578	2.594	2.875	3.000	3.125	3.312
028	1.750	2.531	2.656	2.703	2.719	3.000	3.125	3.250	3.437
030	1.875	2.656	2.781	2.828	2.844	3.125	3.250	3.375	3.562
032	2.000	2.781	2.906	2.953	2.969	3.250	3.375	3.500	3.687
034	2.125	2.906	3.031	3.078	3.094	3.375	3.500	3.625	3.812
036	2.250	3.031	3.156	3.203	3.219	3.500	3.625	3.750	3.937
038	2.375	3.156	3.281	3.328	3.344	3.625	3.750	3.875	4.062
040	2.500	3.281	3.406	3.453	3.469	3.750	3.875	4.000	4.187
042	2.625	3.406	3.531	3.578	3.594	3.875	4.000	4.125	4.312
044	2.750	3.531	3.656	3.703	3.719	4.000	4.125	4.250	4.437
046	2.875	3.656	3.781	3.828	3.844	4.125	4.250	4.375	4.562
048	3.000	3.781	3.906	3.953	3.969	4.250	4.375	4.500	4.687
050	3.125	3.906	4.031	4.078	4.094	4.375	4.500	4.625	4.812
052	3.250	4.031	4.156	4.203	4.219	4.500	4.625	4.750	4.937
054	3.375	4.156	4.281	4.328	4.344	4.625	4.750	4.875	5.062
056	3.500	4.281	4.406	4.453	4.469	4.750	4.875	5.000	5.187
058	3.625	4.406	4.531	4.578	4.594	4.875	5.000	5.125	5.312
060	3.750	4.531	4.656	4.703	4.719	5.000	5.125	5.250	5.437
062	3.875	4.656	4.781	4.828	4.844	5.125	5.250	5.375	5.562
064	4.000	4.781	4.906	4.953	4.969	5.250	5.375	5.500	5.687
066	4.125	4.906	5.031	5.078	5.094	5.375	5.500	5.625	5.812
068	4.250	5.031	5.156	5.203	5.219	5.500	5.625	5.750	5.937
070	4.375	5.156	5.281	5.328	5.344	5.625	5.750	5.875	6.062
072	4.500	5.281	5.406	5.453	5.469	5.750	5.875	6.000	6.187
074	4.625	5.406	5.531	5.578	5.594	5.875	6.000	6.125	6.312
076	4.750	5.531	5.656	5.703	5.719	6.000	6.125	6.250	6.437
078	4.875	5.656	5.781	5.828	5.844	6.125	6.250	6.375	6.562
080	5.000	5.781	5.906	5.953	5.969	6.250	6.375	6.500	6.687
082	5.125	5.906	6.031	6.078	6.094	6.375	6.500	6.625	6.812
084	5.250	6.031	6.156	6.203	6.219	6.500	6.625	6.750	6.937
086	5.375	6.156	6.281	6.328	6.344	6.625	6.750	6.875	7.062
088	5.500	6.281	6.406	6.453	6.469	6.750	6.875	7.000	7.187
090	5.625	6.406	6.531	6.578	6.594	6.875	7.000	7.125	7.312
092	5.750	6.531	6.656	6.703	6.719	7.000	7.125	7.250	7.437
094	5.875	6.656	6.781	6.828	6.844	7.125	7.250	7.375	7.562
096	6.000	6.781	6.906	6.953	6.969	7.250	7.375	7.500	7.687
098	6.125	6.906	7.031	7.078	7.094	7.375	7.500	7.625	7.812
100	6.250	7.031	7.156	7.203	7.219	7.500	7.625	7.750	7.937
102	6.375	7.156	7.281	7.328	7.344	7.625	7.750	7.875	8.062
104	6.500		7.406	7.453	7.469	7.750	7.875	8.000	8.187
106	6.625		7.531	7.578	7.594	7.875	8.000	8.125	8.312
108	6.750		7.656	7.703	7.719	8.000	8.125	8.250	8.437
110	6.875		7.781	7.828	7.844	8.125	8.250	8.375	8.562
112	7.000		7.906	7.953	7.969	8.250	8.375	8.500	8.687
114	7.125		8.031	8.078	8.094	8.375	8.500	8.625	8.812
116	7.250							8.750	8.937
118	7.375							8.875	9.062
120	7.500							9.000	9.187
122	7.625							9.125	9.312
124	7.750							9.250	9.437
126	7.875							9.375	9.562
128	8.000							9.500	9.687

(g) UNASSIGNED DASH NUMBERS SHALL NOT BE USED EXCEPT THOSE DASH NUMBERS INCLUDED BY THE FOLLOWING CODING NOTE: DASH NUMBERS INDICATE PREFERRED GRIP LENGTHS IN .125 INCHEMENTS FOR LENGTHS LONGER THAN .250 INCHES. INTERMEDIATE GRIP LENGTHS BETWEEN THOSE LENGTHS LISTED MAY BE SPECIFIED IN .062 INCREMENTS BY INTERMEDIATE DASH NUMBERS. LENGTH AND GRIP OF A BOLT WITH AN ODD NUMBER FOR THE DASH NUMBER WILL BE .062 GREATER THAN SHOWN FOR THE BOLT WITH THE PRECEDING EVEN DASH NUMBER.

REVIEWER SYMBOLS: NAVY - AS, USAF - 11, ARMY - AV

USER SYMBOLS:

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APPROVED 2 NOV 85 REVISED (A) FOR CHANGES SEE SHEET 1

P A: NAVY - AS; Other Cust: USAF - 11, ARMY - AV

TITLE: BOLT, SELF-LOCKING, 250°F., CRES, 48 KSI Fsu, 80 KSI Ftu, HEX HEAD

MILITARY STANDARD

MS21095

PROCUREMENT SPECIFICATION: MIL-D-6812 & MIL-F-18240

SUPERSEDES (A) MS21095(ASG)

SHEET 3

DD FORM 672-1 (Coordinated)

PLATE NO. 15329

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